

LESSON 6.10

SOLVING MULTI-STEP PROPORTION PROBLEMS



LESSON INTRODUCTION

MA.7.AR.4.5 Solve real-world problems involving proportional relationships.

LEARNING TARGETS

- I can solve multi-step problems involving proportions.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

WORKING TOWARD

MA.8.AR.3 Extend understanding of proportional relationships to two-variable linear equations.

ESSENTIAL QUESTIONS

- How can you use proportional relationships to solve a multi-step ratio problem?

LESSON NARRATIVE

This lesson extends from the previous, building students' understanding and ability to unpack multi-step problems. The lesson includes a Gallery Walk, where students discuss three different contexts and brainstorm various mathematical questions that could be answered using the information given. Students then solve problems within those same three contexts.

COMPONENT SUMMARY

6.10.1 WARM-UP (~5 MIN.)

Video: Cartography, Photogrammetry, and Math

6.10.3 ACTIVITY (10-15 MIN.)

Apply metacognitive thinking skills in exploring approaches to problem-solving in a variety of contexts

6.10.5 WRAP-UP (~5 MIN.)

Demonstrate understanding of solving multi-step problems involving proportional relationships

6.10.2 COLLABORATION (10 MIN.)

In pairs, determine equivalent ratios to solve a problem in context in two different ways

6.10.4 YOUR TURN! (15 MIN.)

Individually solve problems based on the scenarios from the Gallery Walk

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- 6.10.1 Warm-Up Video from the Teacher Area
- Poster paper and sticky notes for 6.10.3 Activity

ADDITIONAL PREPARATION

- Prepare posters around the room with information prompts for 6.10.3 Activity. For smaller groups, consider making two to three copies of each scenario for a total of nine posters, keeping groups to three to four students per group.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may think each problem situation can only have one question that can be solved using that information.
- Students may be thinking additively rather than proportionally (using additive reasoning in place of multiplicative reasoning).
- Students may focus only on the values in the problem and ignore the context of the problem.

THINGS TO CONSIDER

- The last several lessons have involved students thinking proportionally, and it is appropriate for students to use multiple strategies to solve multi-step problems.

LITERACY AND LANGUAGE ROUTINES

Co-Craft Questions and Problems

Starting in 6.10.3 Activity, students are able to use conversation skills to generate, choose (argue for the best one), and improve questions and situations as well as develop meta-awareness of the language used in mathematical questions and problems.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

6.10.3 Activity slows students down so they can decompose a complex problem structure and create plans and procedures to order their steps to solve. Present this activity to students as an opportunity to notice and wonder about what they may miss if they do not slow down to consider the problem context in the future.

DIFFERENTIATE INSTRUCTION

6.10.3 Activity allows students to think independently, write, and then present their reasoning in order to make their thoughts stronger while having the opportunity to listen and understand other students' mathematical reasoning. All students could engage with all scenarios, or groups of students could be strategically assigned to certain scenarios based on their strengths and gaps from previous lessons. Students constructing their own set of questions require a different skill set than analyzing the work and questions of another group.

6.10.1 WARM-UP: MAKING MAPS

DIFFERENTIATE INSTRUCTION

To support students to help direct their attention, have students preview the questions before watching the video.



6.10.1 Warm-Up: Making Maps

Cartographers and photogrammetrists are the professionals who collect and analyze data that they then use to create maps. Cartographers design accessible maps for general use, while photogrammetrists create more detailed maps of Earth's features.



1. What are some of the ways mathematics can be used for map creation?

Mathematics could be used to determine the scale of the map, distances between countries/cities, creating boundary lines, etc.

Answers will vary.

6.10.2 COLLABORATION: THE “RIGHT” WAY

ENRICHMENT

- What is the original ratio given by the recipe?
- What did Lucas change about the recipe?

TEACHER MOVES

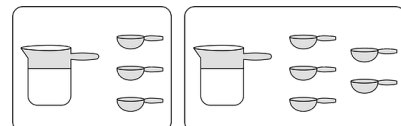
After partners finish questions 1-2, facilitate a whole-class discussion to identify the two different questions that were asked about the same situation and how the questions influenced how they solved the questions differently.



6.10.2 Collaboration: The “Right” Way

Work with your partner to answer the following questions.

Lucas's favorite hot cocoa recipe calls for 1 cup (c) of milk and 3 tablespoons (tbsp) of cocoa. Today, he accidentally added 5 tablespoons of cocoa to 1 cup of milk and it was way too chocolatey.



1. His sister Meredith told him he could just add more milk to fix the flavor. How much milk would he need to add to get the right flavor?

$$\frac{\text{milk}}{\text{cocoa}} \left| \frac{1 \text{ cup}}{3 \text{ Tbsp}} = \frac{?}{5 \text{ Tbsp}} \right. \quad \frac{5}{3} \text{ cup} = 1 \frac{2}{3} \text{ cup}$$

Add $\frac{2}{3}$ cup
of milk

2. His brother Tyler told him he could fix it by adding both milk and cocoa. How much more of each would he need to add to get the right flavor?

$$\frac{\text{milk}}{\text{cocoa}} \left| \frac{1 \text{ cup}}{3 \text{ Tbsp}} = \frac{2 \text{ cup}}{6 \text{ Tbsp}} \right|$$

Add 1 cup of milk and 1 Tbsp of cocoa

3. Which sibling had the correct solution to Lucas's problem? Explain.

Both siblings are correct. They both made equivalent fractions with the hot cocoa recipe.



6.10.3 Activity: Gallery Walk

With your group, read the information given in each scenario. Then, write three different questions that could be answered using the information that was given.

Scenario	Questions That Could Be Answered
Two Maps	<i>How far apart are Windsor and Hawthorne in Map A? What is the scale for Map A? How far apart are Hawthorne and Grandin in Map B?</i>
Favorite Pasta	<i>How many total students took part in the survey? Which pasta dish did students like most? Will lasagna have the least, most, or middle number of students who prefer that dish?</i>
Making Muffins	<i>How many cups of flour are needed for the standard recipe of 12 muffins? What is the ratio of flour to sugar? How many cups of flour do we have?</i>

6.10.3 ACTIVITY: GALLERY WALK

TEACHER MOVES

For larger classes, consider making two copies of each scenario so that there are a smaller number of students at each station in the Gallery Walk at one time.

TEACHER MOVES

Co-Craft Questions and Problems

This activity leans heavily on students building clarity about the problem situation by thinking flexibly about the information given in the problem. Encourage students to come up with three different questions that aren't tightly related to each other. Instruct them to think about the information the problem doesn't provide.

Choose one scenario. Discuss how each question could be answered. Then complete the table.

Question	My Summary of How to Answer the Question
<i>How far apart are Windsor and Hawthorne in Map A?</i>	<i>Map A shows how far Windsor is from Hawthorne in cm.</i>
<i>What is the scale for Map A?</i>	<i>We can use the scale of Map A to determine how far in miles that Windsor and Hawthorne are. I can use a proportion to solve.</i>
<i>How far apart are Hawthorne and Grandin in Map B?</i>	<i>Once I use the proportion from Map A, I can use that distance to make a new proportion to determine what the scale would be for Map B. Once I have the scale for Map B, I can then use that scale to find the distance using how far apart Windsor and Grandin are on the Map B.</i>

TEACHER MOVES

Remind students that they are not actually answering the questions. Rather, they are sharing strategies with each other for how they might approach answering each question with the information given. Consider setting a timer for three to four minutes so that students are focused on discussing how rather than actually trying to work out the answers.

6.10.4 YOUR TURN!

TEACHER MOVES

Encourage students to use the plans from the previous activity to help them solve these problems, when applicable. They already had the chance to make sense of each scenario with their group, so they are encouraged to solve the problems independently.

ENRICHMENT

- How is question 2b similar to question 2a?
- How is it different?

TEACHER MOVES

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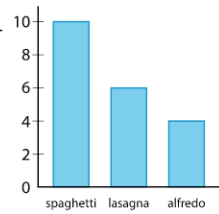
Students could represent their responses in a variety of ways. For example, to find the distance between Windsor and Hawthorne, any of the following proportions could be used:

$$\frac{1 \text{ cm}}{3.5 \text{ mi}} = \frac{2.9 \text{ cm}}{?} \text{ or } \frac{2.9 \text{ cm}}{1 \text{ cm}} = \frac{?}{3.5 \text{ mi}} \text{ or } \frac{3.5 \text{ mi}}{1 \text{ cm}} = \frac{?}{2.9 \text{ cm}}$$



6.10.4 Your Turn!

- Chen surveys a random sample of students about their favorite pasta dish served by the cafeteria and makes a bar graph of the results.



There are 1,256 students who attend the school. Use Chen's data to estimate the number of students at the school whose favorite dish is lasagna.

$$\frac{6}{20} = \frac{l}{1256} \quad l = 376.8$$

$$l = \frac{6 \times 1256}{20} \quad \text{About 377 students prefer lasagna}$$

- Jaxon is baking blueberry muffins with a recipe that calls for 2 cups (c) of flour and $\frac{3}{4}$ cup of sugar to make 12 muffins. Jaxon wants to make as many muffins as possible using 5 cups of flour.

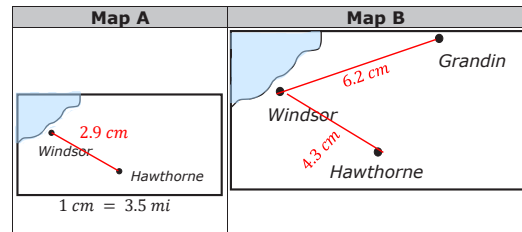
- Exactly how many cups of sugar will Jaxon use if he uses all 5 cups of flour?

$$\frac{\frac{3}{4} \text{ cups}}{2 \text{ cups}} = \frac{?}{5 \text{ cups}} \quad 1\frac{7}{8} \text{ cups}$$

- Exactly how many muffins will Jaxon make if he uses all 5 cups of flour?

$$12 \times 2.5 = 30 \text{ muffins}$$

- The two maps show the same area at two different scales.



Grandin is not on Map A, and Map B does not have a scale.

- Windsor and Hawthorne are 2.9 cm apart on Map A.
- Windsor and Hawthorne are 4.3 cm apart on Map B.
- Windsor and Grandin are 6.2 cm apart on Map B.

What is the distance, in miles, between Windsor and Grandin? Round to the nearest hundredth.

$$\frac{1 \text{ cm}}{2.9 \text{ cm}} = \frac{3.5 \text{ mi}}{?}$$

$$\frac{1 \text{ cm}}{4.3 \text{ cm}} = \frac{?}{10.15 \text{ mi}}$$

$$\frac{2.9 \text{ cm} \times 3.5 \text{ mi}}{1 \text{ cm}} = 10.15 \text{ mi}$$

$$\frac{1 \text{ cm} \times 10.15 \text{ mi}}{4.3 \text{ cm}} = 2.36 \text{ mi}$$

$$\frac{1 \text{ cm}}{6.2 \text{ cm}} = \frac{2.36 \text{ mi}}{x}$$

$$\frac{2.36 \text{ mi} \times 6.2 \text{ cm}}{1 \text{ cm}} \approx 14.63 \text{ miles}$$



6.10.5 Wrap-Up: Ticket Out the Door

- David's car can travel about 25 miles per gallon and his fuel tank holds 12 gallons. He's driving 663 miles on a road trip from Miami, Florida to Atlanta, Georgia. If he starts with a full tank of gas in Miami, how many times will he have to stop for gas on the trip to Atlanta?

$$\frac{25 \text{ mi.}}{\cancel{\text{gal}}} \times 12 \cancel{\text{gal}} = 300 \text{ miles for full tank}$$

$$\frac{1 \text{ tank}}{300 \text{ mi}} = \frac{x}{663 \text{ mi}}$$

$$\frac{1 \text{ tank} \times 663 \cancel{\text{mi}}}{300 \cancel{\text{mi}}} = 2.21 \approx 2 \text{ tanks}$$

If he starts with a full tank, he will have to stop for gas twice on his trip.

6.10.5 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

Ask students to write their final answer in a sentence if this is not a habit for students yet. Encourage them to use the question in the problem to structure their final sentence.